

AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Faculty of Science and Technology · Department of Computer Science

CSC 2210 — Object Oriented Programming 2

Project Assignment 01 — System Design & Report

PROJECT REPORT

MOVIE THEATER MANAGEMENT SYSTEM

Field	Details
Semester	Summer 2025–2026
Section	<u>AA</u>
Group No.	<u>NONE</u>
Domain	Movie Theater / Cinema Ticket Booking
Supervisor	<u>DR.MD.IFTEKHARUL MOBIN</u>

Group Members:

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OTHER LINKS:

GITHUB REPOSITORY LINK	https://github.com/RidwanurRahman/MOVIE-THEATRE-MANAGEMENT-SYSTEM.git	CLICK HERE
SELF EXPLAINED VIDEO LINK	https://drive.google.com/file/d/1vPjpSQC4lwUBZ0BiRRg9GP0mvW6ZX21W/view?usp=sharing	CLICK HERE

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Chapter 1 — Introduction & Case Study

Booking a movie ticket in Bangladesh today usually means calling a theater's box office, standing in a long queue on release day, or hoping a third-party app has bothered to list a small-town cinema at all. Independent multiplexes and single-screen theaters outside the two or three big national chains have almost no online presence: they cannot afford their own booking website, so seats go unsold on slow weekdays while sold-out shows turn away walk-in customers who had no way of checking availability in advance. Meanwhile, movie-goers who do want to plan ahead — compare showtimes across theaters, read reviews before choosing a film, or grab a discounted weekday ticket — have no single place to do it. The Movie Theater Management System solves this by acting as the digital middleman between the people who own theaters and the people who want to watch movies in them, the same way Pathao connects riders with drivers or Daraz connects shoppers with sellers.

The platform has three actors. The Customer is the movie-goer who wants to search for a film, see which theaters are screening it and when, book a specific number of seats, pay online, and later look back at their booking history or leave a rating. The Admin, functioning as a Theater Owner, is the business partner: they run a real cinema hall or multiplex screen and use the platform to list their showtimes, track how many seats remain for each show, see how much they have earned, and run limited-time discount offers to fill empty seats on weekdays. The Super Admin is the platform's own operating company — the party that built and owns the software. They approve new theater owners before those owners can start selling tickets, keep an eye on theaters that are earning consistently poor ratings, moderate reviews that are abusive or fake, and see, across the entire platform, how much revenue and commission has been generated.

The world these three actors share is made of a small number of connected entities. A Theater is a physical cinema business, identified by its name, location, address and the Admin account that owns it. A Show is a single scheduled screening — one movie, on one screen, at one date and time, sold by exactly one theater; every show carries its own title, genre, language, ticket price, total seat count and how many seats remain, so a hugely popular midnight release and a quiet Tuesday matinee of the same film are tracked as two completely separate records. A Cart holds the shows a customer is thinking about booking before they commit to paying, an Order is created the moment they check out, and because one order can contain tickets to more than one show (and a single show is bought by many different orders), an OrderItems table sits between them recording exactly how many seats and at what price were bought in that transaction. A Review ties a customer's star rating and comment to the specific show they watched, and an Offer is a time-boxed percentage discount a theater owner attaches to one of their shows.

Money flows in one direction and splits in two. When a customer pays for tickets at checkout, the full ticket price is charged to their card or mobile wallet. The Movie Theater Management System, as the platform operator, retains a commission percentage from that transaction — the same way a delivery app keeps a cut of every food order — before crediting the remaining balance to the theater owner's earnings, which they can see summarized on their own sales report. The Super Admin's platform-wide dashboard simply aggregates this commission across every theater and every order to show the company how the whole business is performing.

Chapter 2 — Functional Requirements & User Stories

2.1 Functional Requirements

Super Admin

1. Approve or reject new theater owner registrations — new Admin sign-ups sit in "Pending" status until approved.
2. Suspend or permanently delete a theater owner, particularly one with poor customer ratings.
3. View the full list of Admins and Customers, with search and filter by status/role.
4. View a platform-wide sales dashboard showing total bookings, total revenue and commission earned per theater.
5. View a low-rated theaters report listing theaters whose average rating falls below a defined threshold (e.g. < 2.5).
6. Add, edit and delete entries in the master movie-genre/category list.
7. Moderate reviews by deleting abusive or fake reviews.

Admin (Theater Owner)

1. Register a new theater and manage its profile (name, location, address, contact, logo).
2. Perform full CRUD on their own shows/screenings through a DataGridView (add, view, update, delete).
3. View a seat-inventory dashboard showing seats sold, seats remaining, and a low-seat alert when availability drops below a minimum threshold.
4. View an earnings/sales report showing who booked what, on what date and time, at what unit price, and total earnings.
5. Create discount offers/packages with a percentage discount, a start date and an end date.
6. View (read-only) reviews and ratings left on their shows — cannot delete a customer review.
7. Update their own password and profile information.

Customer

1. Sign up for a new account / sign in to an existing account.
2. Browse shows/movies listed by all theaters on the platform.
3. Search for a show by movie name or keyword.
4. Filter shows using dropdowns for at least three criteria: price range, genre, location, language, or date.
5. View full show details and read existing reviews before booking.
6. Add a show to the cart, update the seat quantity, or remove it from the cart.
7. Check out and pay for the cart, generating a bill/invoice.
8. View booking history and reprint a past e-ticket/invoice.
9. Give a 1-5 star rating and write a review for a show they have watched.
10. View special packages and discount offers currently running.
11. Update their own profile information and password.

2.2 User Stories

Super Admin User Stories

As a Super Admin, I can approve or reject new theater owner registrations, so that only legitimate businesses can sell tickets on the platform.

Details: New Admin sign-ups appear in a "Pending Theater Owners" grid. The Super Admin selects a row and clicks Approve or Reject. Approving sets the owner's Status to Active in the Users table and unlocks their dashboard; rejecting sets Status to Rejected and blocks login.

As a Super Admin, I can suspend or delete a theater owner, so that theaters with a poor track record can no longer operate on the platform.

Details: From the Manage Theater Owners grid, the Super Admin clicks Suspend/Delete on a selected owner after a confirmation dialog. Suspending sets Status to Suspended (login blocked, data kept); deleting removes the row after checking for unresolved orders.

As a Super Admin, I can view all users on the platform, so that I can monitor and search the full customer and admin base.

Details: A DataGridView lists every user with name, email, role and status columns. A search box filters by name/email and a dropdown filters by UserType or Status.

As a Super Admin, I can view a platform-wide sales dashboard, so that I can track total orders, total revenue and commission earned per theater.

Details: Summary cards show total bookings, revenue and commission; a bar chart and a per-theater table are computed by joining Orders, OrderItems, Shows and Theaters, filterable by date range.

As a Super Admin, I can view a low-rated shop report, so that I can identify theaters whose service quality is falling below an acceptable standard.

Details: A GROUP BY/HAVING query averages each theater's review ratings and lists theaters with an average below 2.5, alongside review count.

As a Super Admin, I can manage the master list of movie genres, so that theater owners choose from a consistent, clean category list when creating shows.

Details: A CRUD form lets the Super Admin add, rename or delete a genre; deleting a genre still in use by a show is blocked with a validation message.

As a Super Admin, I can moderate reviews, so that abusive or fake reviews do not stay visible on the platform.

Details: A DataGridView lists all reviews; selecting one and clicking Delete Review removes it permanently. Theater owners cannot perform this action.

Admin (Theater Owner) User Stories

As a Theater Owner, I can register my theater and manage its profile, so that customers can find and recognize my business on the platform.

Details: A form collects theater name, location, address, contact and a logo upload. Name and address cannot be empty; on Save a row is inserted into or updated in Theaters linked to the logged-in owner.

As a Theater Owner, I can add a new show, so that customers can discover and book tickets for it.

Details: Clicking Add Show opens a form for movie title, genre (dropdown), language, date, time, screen, price, total seats and a poster. Price and seats must be positive and title cannot be empty, otherwise a red error label appears and Save stays disabled. On Save, a row is inserted into Shows with the logged-in owner's TheaterId.

As a Theater Owner, I can update or delete my own shows, so that I can correct mistakes or remove screenings that are no longer running.

Details: Selecting a row in the show grid and clicking Update reopens the Add Show form pre-filled for editing; Delete asks for confirmation and is blocked if the show already has confirmed bookings.

As a Theater Owner, I can view a seat-inventory dashboard, so that I always know how many seats are sold and how many remain for each show.

Details: A grid lists every show with TotalSeats, seats sold and SeatsAvailable computed live; rows below MinSeatAlert are highlighted red with a warning banner.

As a Theater Owner, I can view an earnings and sales report, so that I know exactly how much revenue each show has generated.

Details: A date-filterable report lists each booking with customer, show, date/time, quantity, unit price and line total, with a SUM-based total at the top.

As a Theater Owner, I can create discount offers on my shows, so that I can fill empty seats on slow days.

Details: A form lets the owner pick a show, a discount percentage, a start date and an end date, validated so StartDate < EndDate and 0-100%, before inserting into Offers.

As a Theater Owner, I can view reviews and ratings left on my shows, so that I can understand customer feedback, but I cannot delete them.

Details: A read-only DataGridView lists reviewer name, rating and comment per show; no delete/edit control is present, since moderation is reserved for the Super Admin.

As a Theater Owner, I can update my own password and profile, so that I can keep my account information current and secure.

Details: A form pre-fills profile fields; changing the password requires entering it twice with a minimum-length check before the Users row is updated.

Customer User Stories

As a Customer, I can sign up and sign in, so that I can access the platform and start booking tickets.

Details: Sign Up validates a unique email and a 6+ character password before inserting a new Users row with UserType Customer. Login checks credentials and routes to Customer Home.

As a Customer, I can browse shows from all theaters, so that I can see everything available to watch nearby.

Details: The Home screen loads all upcoming shows across every theater, pulling MovieTitle, theater name, date/time and price for each entry.

As a Customer, I can search for a show by keyword, so that I can quickly find a specific movie I already have in mind.

Details: A search box filters the show list using a LIKE query on MovieTitle as the customer types or on pressing Search.

As a Customer, I can filter shows using dropdowns, so that I can narrow results down by genre, location and price range at minimum.

Details: Three or more ComboBox dropdowns combine with AND logic in the WHERE clause; selecting All in a dropdown removes that filter.

As a Customer, I can view a show's details and read its reviews, so that I can decide whether it is worth booking before paying.

Details: The details page shows poster, theater, date/time, price and a grid of existing reviews with reviewer name and star rating.

As a Customer, I can add a show to my cart and update the seat quantity, so that I can book more than one ticket at a time.

Details: Add to Cart with a Quantity selector inserts or updates a Cart row; quantity must be positive and cannot exceed SeatsAvailable.

As a Customer, I can check out and pay for my cart, so that I receive confirmed tickets.

Details: Checkout shows cart items, total and a payment method dropdown. On Confirm Payment, one Orders row and one OrderItems row per line are inserted inside a transaction, SeatsAvailable is decremented, and a printable invoice/e-ticket is shown.

As a Customer, I can view my booking/order history, so that I can look back at what I have booked and reprint an invoice if needed.

Details: A grid lists past Orders with date, items and total; selecting a row reopens the invoice layout with a Print button.

As a Customer, I can give a rating and write a review for a show I have watched, so that I can share feedback with other movie-goers.

Details: A 1-5 star selector plus comment box is available from booking history or show details; Submit inserts a Reviews row validated by a CHECK(Rating 1-5) constraint.

As a Customer, I can view special packages and discount offers, so that I can find cheaper tickets during promotional periods.

Details: An Offers screen lists shows with an active Offer (today between StartDate and EndDate) alongside original price, discount percent and calculated discounted price.

As a Customer, I can update my own profile and password, so that I can keep my contact details accurate and my account secure.

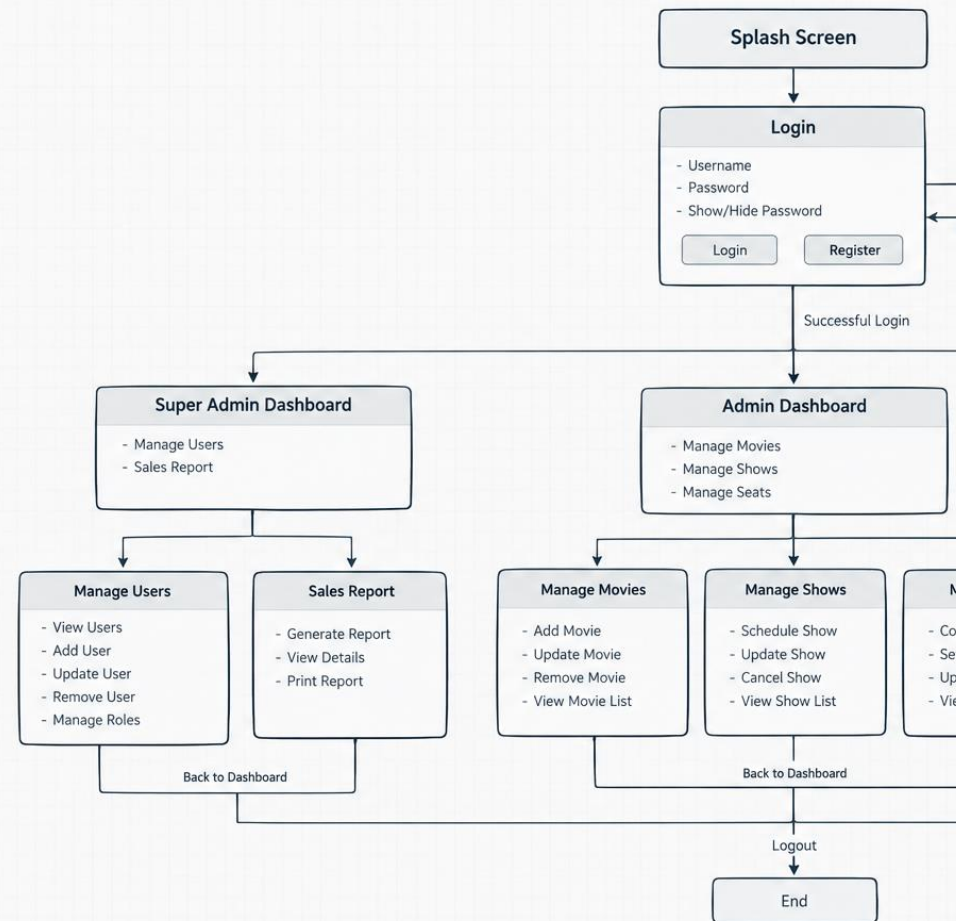
Details: A profile form pre-fills name, phone and address; changing password requires the current password plus a new password entered twice.

Chapter 3 — UI Navigation Diagram

UI NAVIGATION DIAGRAM:

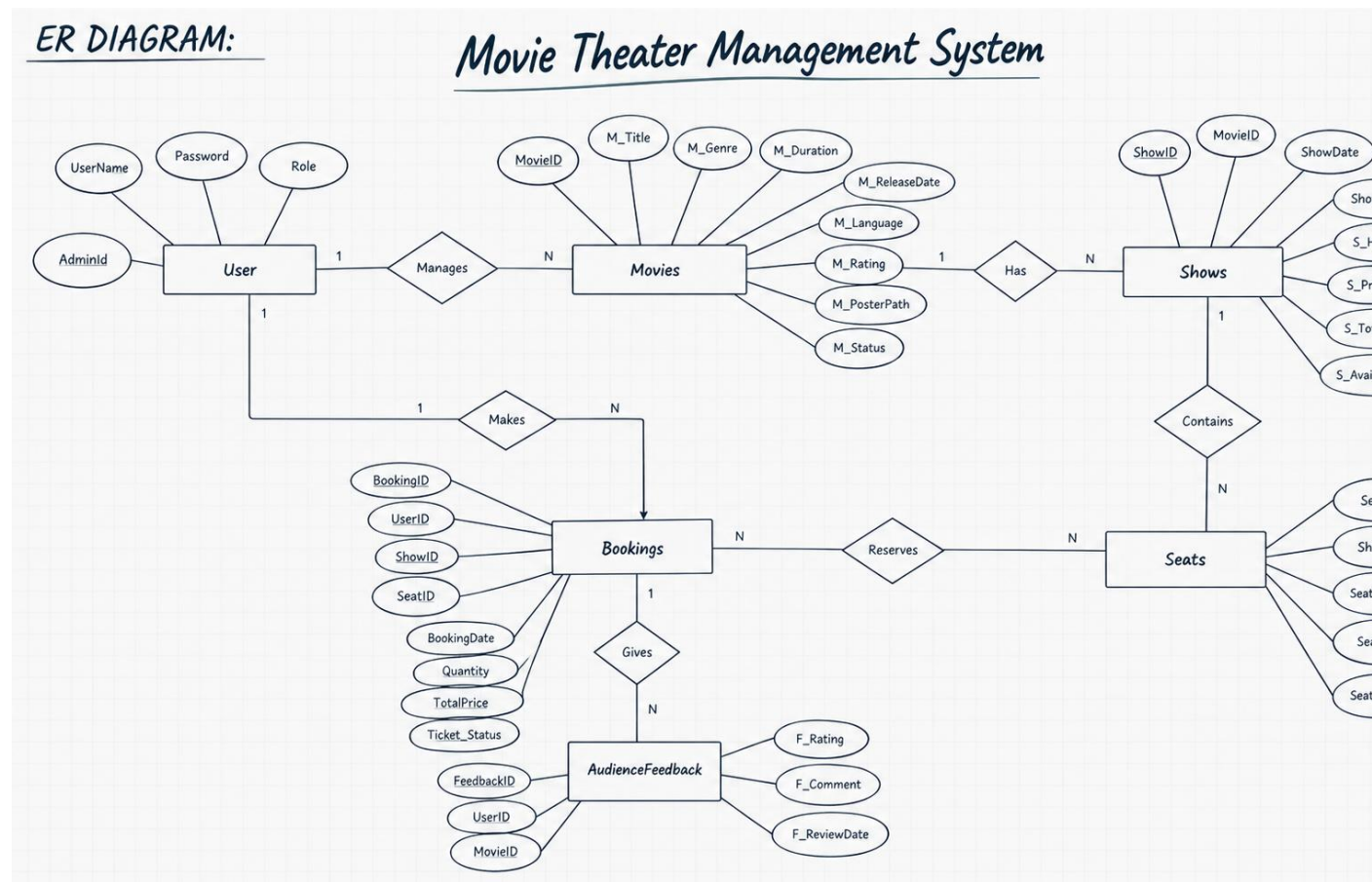
Movie Theater Management

UI Navigation Diagram



Chapter 4 — Database Design

Chapter 4.1 – Er Diagram



Chapter 4.2 — Schema sql Diagram

Movie Theater Management System

Database Schema

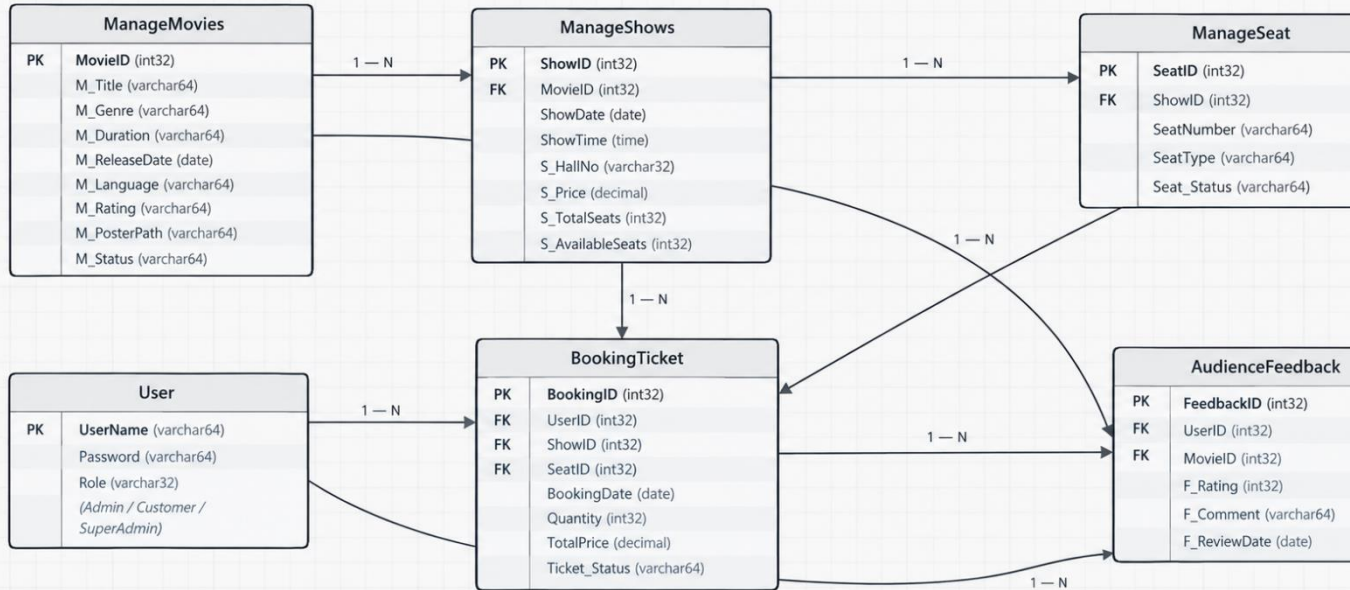


Table Descriptions

Users

Column	Data Type	Constraint	Description
UserId	INT	PK, IDENTITY	Unique user identifier
FullName	VARCHAR(100)	NOT NULL	User's full name
Email	VARCHAR(100)	NOT NULL, UNIQUE	Login email
Password	VARCHAR(255)	NOT NULL	Hashed password
Phone	VARCHAR(20)	NOT NULL	Contact number
Address	VARCHAR(200)		Mailing/city address
UserType	VARCHAR(20)	NOT NULL, CHECK	SuperAdmin / Admin / Customer
Status	VARCHAR(20)	NOT NULL, CHECK	Pending / Active / Suspended
CreatedAt	DATETIME	NOT NULL	Registration timestamp

Genres

Column	Data Type	Constraint	Description
GenreId	INT	PK, IDENTITY	Unique genre identifier
GenreName	VARCHAR(50)	NOT NULL, UNIQUE	e.g. Action, Drama
Description	VARCHAR(200)		Short description

Theaters

Column	Data Type	Constraint	Description
TheaterId	INT	PK, IDENTITY	Unique theater identifier
OwnerId	INT	FK -> Users(UserId)	Owning Admin account
TheaterName	VARCHAR(100)	NOT NULL	Business name
Location	VARCHAR(100)	NOT NULL	Area/city
Address	VARCHAR(200)	NOT NULL	Full street address
Contact	VARCHAR(20)		Box-office phone number
Status	VARCHAR(20)	NOT NULL, CHECK	Pending / Active / Suspended

Shows

Column	Data Type	Constraint	Description
ShowId	INT	PK, IDENTITY	Unique show identifier
TheaterId	INT	FK -> Theaters	Owning theater
GenreId	INT	FK -> Genres	Movie genre
MovieTitle	VARCHAR(150)	NOT NULL	Movie name
Language	VARCHAR(30)	NOT NULL	Audio language
ShowDate	DATE	NOT NULL	Screening date
ShowTime	TIME	NOT NULL	Screening time
ScreenNo	VARCHAR(10)	NOT NULL	Hall/screen number
Price	DECIMAL(10,2)	NOT NULL, CHECK>0	Ticket price
TotalSeats	INT	NOT NULL, CHECK>0	Total hall capacity
SeatsAvailable	INT	NOT NULL	Seats currently unsold
MinSeatAlert	INT	NOT NULL	Low-seat alert threshold
Description	VARCHAR(300)		Short synopsis
PosterPath	VARCHAR(255)		Poster image path

Cart

Column	Data Type	Constraint	Description
CartId	INT	PK, IDENTITY	Unique cart-line identifier

CustomerId	INT	FK -> Users	Owning customer
ShowId	INT	FK -> Shows	Selected show
Quantity	INT	NOT NULL, CHECK>0	Seats requested
AddedDate	DATETIME	NOT NULL	When added to cart

Orders

Column	Data Type	Constraint	Description
OrderId	INT	PK, IDENTITY	Unique order identifier
CustomerId	INT	FK -> Users	Booking customer
OrderDate	DATETIME	NOT NULL	Checkout timestamp
TotalAmount	DECIMAL(10,2)	NOT NULL	Full order total
PaymentMethod	VARCHAR(30)	NOT NULL	Card / Mobile Banking / etc.
Status	VARCHAR(20)	NOT NULL, CHECK	Confirmed / Cancelled

OrderItems (junction table)

Column	Data Type	Constraint	Description
OrderItemId	INT	PK, IDENTITY	Unique line-item identifier
OrderId	INT	FK -> Orders	Parent order
ShowId	INT	FK -> Shows	Booked show
Quantity	INT	NOT NULL, CHECK>0	Seats booked in this line
UnitPrice	DECIMAL(10,2)	NOT NULL	Price per seat at booking time
Subtotal	DECIMAL(10,2)	NOT NULL	Quantity x UnitPrice

Reviews

Column	Data Type	Constraint	Description
ReviewId	INT	PK, IDENTITY	Unique review identifier
CustomerId	INT	FK -> Users	Reviewer
ShowId	INT	FK -> Shows	Reviewed show
Rating	INT	NOT NULL, CHECK 1-5	Star rating
Comment	VARCHAR(500)		Written feedback
ReviewDate	DATETIME	NOT NULL	Submission timestamp

Offers

Column	Data Type	Constraint	Description
OfferId	INT	PK, IDENTITY	Unique offer identifier
ShowId	INT	FK -> Shows	Discounted show
DiscountPercent	DECIMAL(5,2)	NOT NULL, CHECK 0-100	Percentage off
StartDate	DATE	NOT NULL	Offer start
EndDate	DATE	NOT NULL	Offer end

Normalisation Justification (3NF)

Every table has a single-column surrogate primary key, and every non-key column depends only on that key (2NF is satisfied since there are no composite keys with partial dependencies). No non-key column depends on another non-key column: Subtotal in OrderItems is stored for query performance, but it is functionally derived only from Quantity and UnitPrice within the same row, not from another table, so it introduces no transitive dependency. The OrderItems junction table is the key 3NF decision — instead of storing a comma-separated list

of shows inside Orders, which would violate 1NF, each booked show gets its own row linked back to both Orders and Shows. GenreName lives only in Genres and is referenced by GenreId from Shows rather than repeated as text on every show, removing update anomalies. Theaters.OwnerId and Shows.TheaterId enforce data isolation directly in the schema: every query the Admin dashboard runs is filtered by the logged-in owner's TheaterId, so one Theater Owner can never see another owner's shows, orders or earnings.

Chapter 5 — SQL Queries

1. LOGIN

Verifies the entered email and password against Users and returns the UserType so the Login form can route to the correct dashboard.

2-3. FILTER

Filters the show list by price range, and separately by genre plus seat availability, backing the Customer's dropdown filters.

4. SEARCH

Uses a LIKE pattern on MovieTitle to support keyword search from the Browse Shows screen.

5. CART

Adds a row to Cart, deletes a row from Cart, and computes each cart line's subtotal (Quantity x Price) for the Cart screen.

6. CHECKOUT

Wraps the booking in a transaction: inserts one Orders row, one OrderItems row, and decrements SeatsAvailable, so partial failures cannot leave inconsistent seat counts.

7. THEATER EARNINGS

Joins OrderItems, Shows and Theaters and uses SUM with GROUP BY to total revenue per theater, powering both the Admin earnings report and the Super Admin platform dashboard.

8. LOW SEAT ALERT

Compares SeatsAvailable against MinSeatAlert to drive the red warning banner on the Admin inventory dashboard.

9. REVIEWS

Joins Reviews with Users to display the reviewer's name alongside their rating and comment on the Show Details screen.

10. LOW-RATED THEATERS

Uses AVG with GROUP BY and HAVING to find theaters whose average rating across all their shows falls below 2.5.

11. SUSPEND/DELETE OWNER

Updates a theater owner's Status to Suspended (or deletes the row) from the Super Admin's Manage Theater Owners screen.

12. OFFERS

Joins Offers with Shows and calculates the discounted price for any offer whose date range includes today, for the Customer's Offers screen.

Full Script Excerpt (CREATE TABLE — Shows)

```
CREATE TABLE Shows (  
  ShowId          INT IDENTITY(1,1) PRIMARY KEY,  
  TheaterId       INT NOT NULL FOREIGN KEY REFERENCES Theaters(TheaterId),  
  GenreId         INT NOT NULL FOREIGN KEY REFERENCES Genres(GenreId),  
  MovieTitle      VARCHAR(150) NOT NULL,  
  Language        VARCHAR(30)  NOT NULL,
```

```
ShowDate      DATE          NOT NULL,
ShowTime      TIME          NOT NULL,
ScreenNo      VARCHAR(10)   NOT NULL,
Price         DECIMAL(10,2) NOT NULL CHECK (Price > 0),
TotalSeats    INT NOT NULL CHECK (TotalSeats > 0),
SeatsAvailable INT NOT NULL,
MinSeatAlert  INT NOT NULL DEFAULT 10,
Description   VARCHAR(300),
PosterPath    VARCHAR(255)
);
```

The complete schema.sql file (all 9 CREATE TABLE statements, sample data, and all 12 feature queries) is available in the repository under /database/schema.sql.

Chapter 6 — User Interface Design

All 12 required form mockups follow a consistent colour scheme (navy for Super Admin/entry screens, orange for Admin, green for Customer), a visible title bar on every form, and a Back or Logout control so no screen is a dead end. Data lists use a DataGridView rather than a plain list box, and filters use ComboBox dropdowns rather than free-text boxes.

Login



Sign Up / Registration

X

Registration

UserName

Password

Role

Register

Super Admin Dashboard



Admin dashboard



Audience dash



Manage users

Manage Users

UserName

Password

Role

Approve

Delete

Update

Clear

Search

Manage Users Dashboard

	UserName	Password	Role	Status
▶	A001	1234	Admin	Approved
	C001	1234	Audience	Approved
	C002	1234	Customer	Approved

Manage movies

Manage Movies

MovieID

M005

M_Title

M_Genre

M_Duration

_:

M_ReleaseDate

2026-09-07

M_Language

M_Rating

M_Status

	MovieID	M_Title	M_Genre	M_Duration	M_ReleaseDate	M_Language
▶	M001	Dear X	Thriller	02:00	2026-09-25	Korean
	M002	Titanic	Romantic	02:00	2026-09-25	English
	M003	3 Idiots	Adventure	02:00	2026-09-26	Hindi
	M004	SpiderMan	Adventure	02:00	2026-09-25	English

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Sales Report

Total Revenue

8500.00

Total Confirmed Bookings

Total Tickets Sold

9

Average Booking Value

2125.00

Search

Sales Report Dashboard

	Booking ID	User ID	Show ID	Movie	Seat ID	Booking Date	Quantity	Total Price	Status
▶	BI003	C001	Sh004	Titanic	S004	2026-09-05	3	4500	Confirmed
	BI004	C003	Sh003	3 Idiots	S003	2026-09-05	2	1000	Confirmed
	BI001	C001	Sh001	Dear X	S001	2026-09-04	1	1500	Confirmed

Manage Shows

Manage Shows

ShowID

Sh005

MovieID

ShowDate

2026-09-07

ShowTime

5:04:55 PM

S_HallNo

S_Price

S_TotalSeats

S_AvailableSeats

0

Add

Delete

Update

Movie Data

	MovieID	M_Title	M_Genre	M_Duration
▶	M001	Dear X	Thriller	02:00
	M002	Titanic	Romantic	02:00
	M003	3 Idiots	Adventure	02:00
	M004	SpiderMan	Adventure	02:00

Shows Data

	ShowID	MovieID	ShowDate	ShowTime
▶	Sh001	M001	2026-09-25	20:00:00
	Sh002	M002	2026-09-25	21:00:00
	Sh003	M003	2026-09-05	00:42:00
	Sh004	M002	2026-09-25	20:00:00

Manage seat

Manage Seat

SeatID

S005

ShowID

SeatNumber

SeatType

Seat_Status

Available

Add

Delete

Update

Clear

Show Details

	ShowID	MovielD	ShowDate	Sho
►	Sh001	M001	2026-09-25	20:00
	Sh002	M002	2026-09-25	21:00
	Sh003	M003	2026-09-05	00:42

Seat Details

	SeatID	ShowID	
►	S001	Sh001	1
	S002	Sh002	7
	S003	Sh003	8
	S004	Sh004	1

Booking ticket

BookingID

BI005

UserID

C001

ShowID

SeatID

BookingDate

9/ 7/2026

Quantity

1

TotalPrice

0

Ticket_Status

Confirmed

Add

Update

Delete

Clear

	ShowID	Movie
▶	Sh003	M003
	Sh004	M002
	Sh001	M001

	BookingID	User
▶	BI003	C001
	BI002	C001
	BI001	C001

Feedback

Audience_Feedback

FeedbackID

F003

UserID

C001

MovieID

F_Rating

F_Comment

F_ReviewDate

9/ 7/2026

Add

Delete

Update

Clear

Booking Table

	BookingID	UserID	ShowID	SeatID	MovieID
▶	BI003	C001	Sh004	S004	M002
	BI002	C001	Sh002	S002	M002
	BI001	C001	Sh001	S001	M001

Feedback Table

	FeedbackID	UserID	MovieID	MovieTitle
▶	F001	C001	M002	Titani

Chapter 7 — Work Distribution

Name	ID	Contribution
MUHAMMAD RIDANUR RAHMAN	23-55300-3	Case study, functional requirements, user stories
Alone	^	Database design, SQL schema diagram, schema.sql
Alone	^	UI navigation diagram, ER diagram
Alone	^	UI mockups/form designs, README assembly, PDF report

Chapter 8 — Conclusion & Future Work

This report has produced the complete design blueprint for the Movie Theater Management System: a real-world case study, functional requirements and user stories for all three roles, a UI navigation diagram, a normalised database schema with an accompanying SQL script, and mockups for all twelve required forms. Together these deliverables prove that the requirements, the navigation flow, and the database agree with each other — every feature listed in Chapter 2 is reachable through a form shown in Chapter 3 and is backed by a table and a query shown in Chapters 4 and 5.

In the coding phase that follows, the group will implement this design as a working C# Windows Forms application: building the login and role-based routing first, then each dashboard's DataGridView-driven CRUD screens, and finally the customer-facing booking and payment flow. Planned enhancements beyond the base requirements include seat-map selection (choosing individual seat numbers rather than just a quantity), integration with a real payment gateway such as bKash or a card processor, and automated email/SMS notifications for booking confirmations and low-seat alerts to theater owners.

The group anticipates two main challenges during implementation. The first is concurrency: multiple customers may try to book the last few seats of a popular show at the same time, so the checkout transaction must correctly lock or re-check `SeatsAvailable` to avoid overselling. The second is enforcing the data-isolation requirement strictly at the query layer — every single query issued from an Admin's screens must be filtered by that Admin's own `TheaterId`, since a single missed `WHERE` clause would let one theater owner see another owner's private sales data.